Xinyu Shi

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EDUCATION

Xiamen University **Xiamen, China**
B.Eng. in Software Engineering *Sept. 2016 - Jul. 2020*

- Thesis: Multi-dimensional Interaction Based on Sensing Technique in Mixed Reality.
- GPA: 89/100 (3.71/4.0) Rank: 7/127 (Top 5%)

INTERNSHIPS & EXPERIENCES

Alibaba DAMO Academy **Hangzhou, China**
Research Intern, Artificial Intelligence Center - Visual Computing Lab *Jul. 2020 - present*

- Mentor: Dr. Tao Yang
- Project: Image Super Resolution based on Domain Translation

Dartmouth College **Hanover, NH, USA**
Research Intern, Human-Computer Interaction Lab *Jan. 2020 - Apr. 2020*

- Supervisor: Prof. Xing-Dong Yang
- Project: CircuitLab: A System for Aiding Remote Electronics Labs

Research Intern, Dartmouth Reality and Robotics Lab *Sept. 2019 - Dec. 2019*

- Supervisor: Prof. Xia Zhou
- Project: Teaching American Sign Language in Mixed Reality

Xiamen University **Xiamen, China**
Research Assistant, Human+ Lab *Oct. 2017 - Aug. 2019*

- Supervisor: Prof. Shihui Guo
- HCI research projects: AR/VR/MR, wearable technique, gesture interaction, etc.

Cambridge University **Cambridge, UK**
Visiting Student, Artificial Intelligence Program *Jan. 2019 - Feb. 2019*

- Best Individuals Award for Top 5%
- Group Project: Reinforcement Learning Application in Virtual Avatar Motion Control

PUBLICATIONS

Published

1. **Xinyu Shi**, Junjun Pan, Zeyong Hu, Juncong Lin, Shihui Guo, Minghong Liao, Ye Pan, and Ligang Liu. Accurate and Fast Classification of Foot Gestures for Virtual Locomotion. *International Symposium on Mixed and Augmented Reality (ISMAR), 2019.*
2. Qijia Shao, Amy Sniffen, Julien Blanchet, Megan Elizabeth Hillis, **Xinyu Shi**, Themistoklis Haris, Jason Liu, Lorna Quandt, James Mahoney, David J.M.Kraemer, Xia Zhou, and Devin Balkcom. Teaching American Sign Language in Mixed Reality. Submitted to *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT), 2020. (In press)*
3. Shihui Guo, Jubo Yu, **Xinyu Shi**, Hongran Wang, Feibin Xie, Xing Gao, and Min Jiang. Droplet-transmitted Infection Risk Ranking based on Close Proximity Interactions. *Frontiers in Neurorobotics 13 (2020): 113.*
4. Hongran Wang, Shihui Guo and **Xinyu Shi**. MagicGlove: Estimation for Flexion Angle of Fingers. *Proceedings of the 2019 IEEE Symposium Series on Computational Intelligence (SSCI), 2019.*

Manuscripts

1. Ruei-Che Chang, Te-Yen Wu, Zheer Xu, **Xinyu Shi**, Xueguang Ma, Teddy Seyed and Xing-Dong Yang. CircuitLab: A System for Aiding Remote Electronics Labs. *Submitted to UIST 2020.*

RESEARCH EXPERIENCES

Image Super Resolution based on Domain Translation **Alibaba DAMO Academy**
Mentored by Dr. Tao Yang *Jul. 2020 - present*

- Exploited Generative models (GAN-based and VAE-based) to translate the noisy low-resolution images to clean high-resolution images.
- Manipulated the latent space to achieve the multi-domain transfer in an unsupervised manner.
- Improved the perceptual quality in the real-world deployment setting compared with state-of-the-art methods.

CircuitLab: A System for Aiding Remote Electronics Labs **Dartmouth College**
Supervised by Prof. Xing-Dong Yang *Jan. 2020 - Apr. 2020*

- Augmented existing video conferencing tools with a system continuously provided remote students with context-sensitive learning assistance to better facilitate remote electronics labs.
- Built a 3D printed station holding the smartphone and the breadboard in place to ensure a stable view of the circuit while also allowing the smartphone to sense circuit components (using computer vision: Mask-RCNN).
- Ran a deployment study with an instructor mentoring nine remote students that were constructing a breadboard circuit and our system are proven to be effective.

Teaching American Sign Language Using Mixed Reality **Dartmouth College**
Supervised by Prof. Xia Zhou *Sept. 2019 - Dec. 2019*

- Leveraged the flex and force sensor-based glove system facilitated with ARTag fiducial system to acquire learners' handshape and motion trajectory.
- Allowed learners to perceive their own hands in the mixed-reality environment with the first- and third-person views for motion demonstration performed by the virtual teacher.
- Provided real-time feedback to correct learners' mistakes in practice tailored to the specific learner.

Accurate and Fast Classification of Foot Gestures for Virtual Locomotion **Xiamen University**
Supervised by Prof. Shihui Guo *Oct. 2018 - Apr. 2019*

- Exploited smart insoles for mapping the locomotion in the real world to that in virtual environments.
- Proposed a novel Human Action Recognition method, Dual-Check Till Consensus (DCTC), improved the conventional LSTM by applying the mechanism of probability-based decision-making.
- Reduced the latency of gesture recognition from 2 seconds to 0.5 seconds and increased the accuracy from 85% to over 97%, in comparison with LSTM, HMM, DTW&KNN.

Infection Risk Ranking based on Close Proximity Interactions **Xiamen University**
Supervised by Prof. Shihui Guo *Apr. 2018 - Sept. 2018*

- Aimed to identify people potentially infected by droplet-transmitted diseases from the low-resolution surveillance videos according to the distance-based relationship and body interactions.
- Re-identified the diagnosed patient in non-overlapping monitor cameras using the ResNet-50.
- Determined the degree of infection risks according to 3D distances of their skeleton keypoints while depth information in the 2D image retrieved by depth estimation algorithm using CNN.
- Evaluated the proposed framework in real-world environments, consistent prediction results as human expert observers received.

TECHNICAL STRENGTHS

Programming Languages: Python, C/C++, Java

Development: HTML/CSS, MySQL, Java Spring

Others: PyTorch, Tensorflow, LaTeX

SELECTED HONORS & AWARDS

2020 Honorable Undergraduate Thesis, Xiamen University.

2019 Best Innovative Project at Innovation and Entrepreneurship Conference (Top 1%, 10000 RMB).

2019 Anniversary Scholarship, Xiamen University (Top 3%, 5000 RMB).

2019 Scholarship of Artificial Intelligence Program, Cambridge University (Top 5%, £500).

2019 First-class Scholarship of Academic Achievements, Xiamen University (Top 20%, 1000 RMB).